n-truncation of a pushout cocone is again a pushout among n-types

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'reflectcommutespushout' (source lines 984-989). The proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context carries the author's own equation eq:conetrunc (source lines 978-980) because the proof's first alignment step quotes it by number; retaining it makes that step checkable inside the package. The equation eq:trunc-htpy quoted in that same sentence, and the equation eq:composeconefunc and lemma thm:conemap-funct which sit further back in the same section, are recorded as source references instead.

Editorial note. Theorem-like environments and numbered equations share a single counter in this book (its macro file sets \let\c@equation\c@thm), so the retained counter value 10 lets the retained equation print as 7.4.11 and the counted theorem as Theorem 7.4.12, matching the original numbering exactly.

Editorial note. The proof body contains a local definition '\def\reflect(#1){\trunc n{#1}}' inside a group; it is author notation for the n-truncation and is retained verbatim.

Editorial note. No retained line contains a citation, a figure, an included file or a non-ASCII character, so no bibliography and no asset is declared.

Editorial scope. One original result of the HoTT Book, counted once: n-truncation of a pushout cocone is again a pushout among n-types.

Allowed standard prerequisites (editorial).

- Spans, cocones and the universal property of a pushout. For a span D = (A <-f- C -g-> B) and a type E, a cocone under D with vertex E is a triple (i,j,h) with i : A -> E, j : B -> E and h : Prd{c:C} i(f(c)) = j(g(c)); a pushout of D is a cocone that is initial, i.e. the induced map (D-pushout -> E) -> cocone(D,E) is an equivalence.. The theorem is stated entirely in the language of cocones and pushouts, and the proof manipulates the two cocone types that appear in the square at source lines 994-1003.
- Truncation as a higher inductive type, its recursion and its naturality. For n >= -1 the n-truncation trunc n A is generated by tproj n : A -> trunc n A together with hub and spoke constructors; recursion into an n-type E is by a map on the points, and the naturality homotopies nat^f_n : trunc n

f (tproj n a) = tproj n (f a) are the book's equation 7.3.4.. The proof uses that a map trunc n D -> E into an n-type E is the same as a map D -> E (source line 1005), and it invokes the naturality homotopies explicitly when defining the lowest horizontal arrow (source lines 1028-1034).

- Function extensionality and the two-out-of-three property for equivalences. happly is quasi-inverse to funext, so a homotopy between functions gives a path of functions and conversely; and in a commuting square of types, if two of the three maps are equivalences then so is the third.. The middle horizontal arrow is an equivalence only because both lower vertical arrows are happly (source lines 1017-1023), and the argument applies two-out-of-three twice (source lines 1005-1006 and 1037).

Source reference. Original lemma 7.3.5: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1905–1905. Pointer to the original naturality statement for truncation used just above the retained span to define the truncated cocone.

Source reference. Original equation 7.3.6: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1328–1328. Original equation 7.3.6, the additional condition satisfied by the truncated cocone. The retained context quotes it once to derive the identity eq:conetrunc; it is a cited equation, not part of the counted theorem's own argument.

Source reference. Original lemma 7.4.10: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1752–1752. Original Lemma 7.4.10, the functoriality of the cocone action under a map of spans. The retained proof uses it only to justify the tag on one definitional alignment step (source line 1011); it is not the load-bearing step, which is the two-out-of-three diagram chase retained in full at source lines 1005-1061.

Source reference. Original equation 7.4.4: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1216–1216. Original equation 7.4.4, the functoriality identity for composition of cocones. It is used inside the retained proof as a bookkeeping tag (source line 1015) for a reassociation that is checked within the retained proof itself.

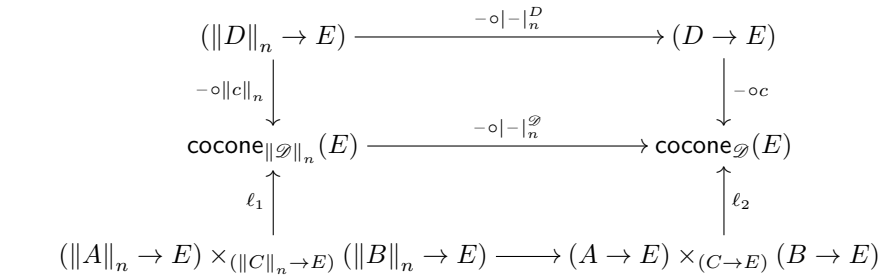
Finally, note that since we defined $\|c\|_n : \mathsf{cocone}_{\|\mathscr{D}\|_n}(\|D\|_n)$ using original lemma 7.3.5, the additional condition (7.3.6) implies

$$|-|_n^D \circ c = \|c\|_n \circ |-|_n^{\mathscr{D}}. \tag{7.4.11}$$

for any $c : \mathsf{cocone}_{\mathscr{D}}(D)$. Now we can prove our desired theorem.

Theorem 7.4.12. *Let $\mathscr{D}$ be a span and (D, c) its pushout. Then $(\|D\|_n, \|c\|_n)$ is a pushout of $\|\mathscr{D}\|_n$ in n-types.*

Proof. Let E be an n-type, and consider the following diagram:



The upper horizontal arrow is an equivalence since E is an n-type, while $- \circ c$ is an equivalence since c is a pushout cocone. Thus, by the 2-out-of-3 property, to show that $- \circ \|c\|_n$ is an equivalence, it will suffice to show that the upper square commutes and that the middle horizontal arrow is an equivalence. To see that the upper square commutes, let $t : \|D\|_n \to E$; then

$$\begin{aligned}
\big(t \circ \|c\|_n\big) \circ |-|_n^{\mathscr{D}} &= t \circ \big(\|c\|_n \circ |-|_n^{\mathscr{D}}\big) && \text{(by original lemma 7.4.10)} \\
&= t \circ \big(|-|_n^D \circ c\big) && \text{(by (7.4.11))} \\
&= \big(t \circ |-|_n^D\big) \circ c. && \text{(by (7.4.4))}
\end{aligned}$$

To show that the middle horizontal arrow is an equivalence, consider the lower square. The two lower vertical arrows are simply applications of happly:

$$\begin{aligned}
\ell_1(i, j, p) &:\equiv (i, j, \mathsf{happly}(p)) \\
\ell_2(i, j, p) &:\equiv (i, j, \mathsf{happly}(p))
\end{aligned}$$

and hence are equivalences by function extensionality. The lowest horizontal arrow is defined by

$$(i, j, p) \mapsto \big(i \circ |-|_n^A,\ \ j \circ |-|_n^B,\ \ q\big)$$

where q is the composite

$$\begin{aligned}
i \circ |-|_n^A \circ f &= i \circ \|f\|_n \circ |-|_n^C && \text{(by } \mathsf{funext}(\lambda z.\, \mathsf{ap}_i(\mathsf{nat}_n^f(z)))\text{)} \\
&= j \circ \|g\|_n \circ |-|_n^C && \text{(by } \mathsf{ap}_{-\circ|-|_n^C}(p)\text{)} \\
&= j \circ |-|_n^B \circ g. && \text{(by } \mathsf{funext}(\lambda z.\, \mathsf{ap}_j(\mathsf{nat}_n^g(z)))\text{)}
\end{aligned}$$

This is an equivalence, because it is induced by an equivalence of cospans. Thus, by 2-out-of-3, it will suffice to show that the lower square commutes. But the two composites around the lower square agree definitionally on the first two components, so it suffices to show that for (i, j, p) in the lower left corner and $z : C$, the path

$$\mathsf{happly}(q, z) : i(|f(z)|_n) = j(|g(z)|_n)$$

(with q as above) is equal to the composite

$$\begin{aligned}
i(|f(z)|_n) &= i(\|f\|_n(|z|_n)) && \text{(by } \mathsf{ap}_i(\mathsf{nat}_n^f(z))\text{)} \\
&= j(\|g\|_n(|z|_n)) && \text{(by } \mathsf{happly}(p, |z|_n)\text{)} \\
&= j(|g(z)|_n). && \text{(by } \mathsf{ap}_j(\mathsf{nat}_n^g(z))\text{)}
\end{aligned}$$

However, since happly is functorial, it suffices to check equality for the three component paths:

$$\begin{aligned}
\mathsf{happly}(\mathsf{funext}(\lambda z.\, \mathsf{ap}_i(\mathsf{nat}^f_n(z))), z) &= \mathsf{ap}_i(\mathsf{nat}^f_n(z)) \\
\mathsf{happly}(\mathsf{ap}_{-\circ|-|^C_n}(p), z) &= \mathsf{happly}(p, |z|_n) \\
\mathsf{happly}(\mathsf{funext}(\lambda z.\, \mathsf{ap}_j(\mathsf{nat}^g_n(z))), z) &= \mathsf{ap}_j(\mathsf{nat}^g_n(z)).
\end{aligned}$$

The first and third of these are just the fact that happly is quasi-inverse to funext, while the second is an easy general lemma about happly and precomposition. $\square$